INPUT-OUTPUT CHANNELS

Vision

Hearing

Touch

Movement

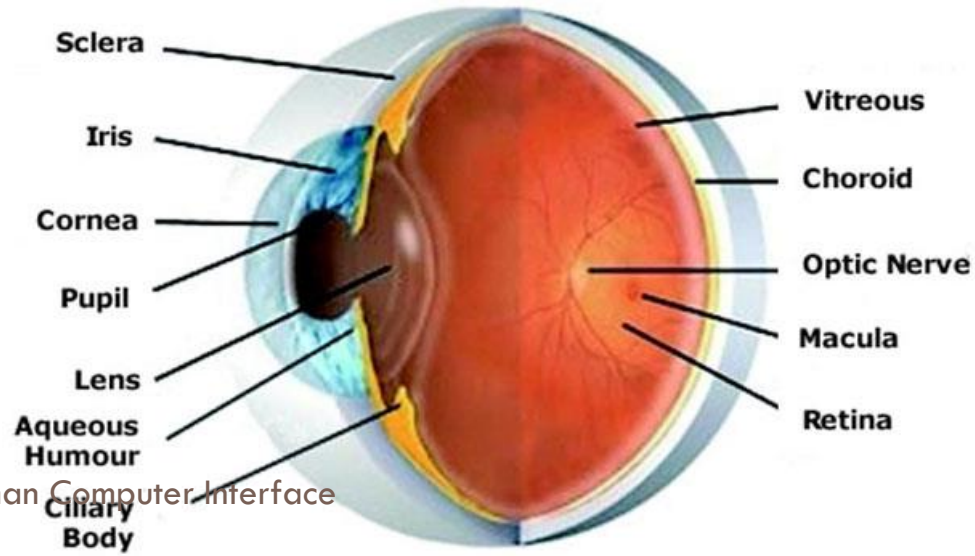
1.Human vision

Two stages in vision

- physical reception of stimulus
- processing and interpretation of stimulus

The Eye - physical reception

- mechanism for receiving light and transforming it into electrical energy
- light reflects from objects
- images are focused upside-down on retina
- retina contains rods for low light vision and cones for colour vision
- ganglion cells (brain!) detect pattern and movement



Interpreting the signal

- Size and depth
  - visual angle indicates how much of view object occupies

    (relates to size and distance from eye)
  - visual acuity is ability to perceive detail (limited)
  - familiar objects perceived as constant size

    (in spite of changes in visual angle when far away)
  - cues like overlapping help perception of size and depth

Interpreting the signal (cont)

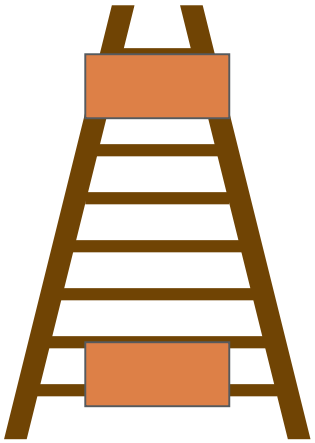
- Brightness
  - subjective reaction to levels of light
  - affected by luminance of object
  - measured by just noticeable difference
  - visual acuity increases with luminance as does flicker
- Colour
  - made up of hue, intensity, saturation
  - cones sensitive to colour wavelengths
  - blue acuity is lowest
  - 8% males and 1% females colour blind

Interpreting the signal (cont)

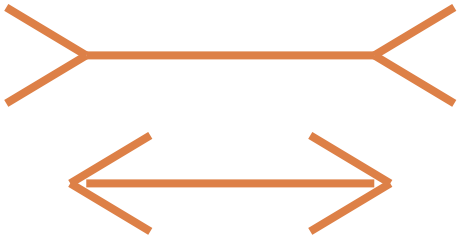
- The visual system compensates for:
  - movement
  - changes in luminance.
- Context is used to resolve ambiguity
- Optical illusions sometimes occur due to over compensation

Optical Illusions

- "These illusion demonstrate that our perception of size is not completely reliable"



the Ponzo illusion



the Muller Lyer illusion

Reading

- Several stages:
  - visual pattern perceived
  - decoded using internal representation of language
  - interpreted using knowledge of syntax, semantics, pragmatics

- Reading involves saccades and fixations
- Perception occurs during fixations
- **In design Focus:**
  - Word shape is important to recognition
  - Negative contrast improves reading from computer screen

2.Hearing

- Provides information about environment:

  distances, directions, objects etc.

- Physical apparatus:
  - outer ear – protects inner and amplifies sound
  - middle ear – transmits sound waves as vibrations to inner ear
  - inner ear – chemical transmitters are released and cause impulses in auditory nerve
- Sound
  - pitch – sound frequency
  - loudness – amplitude
  - timbre – type or quality

Hearing (cont)

- Humans can hear frequencies from 20Hz to 15kHz
  - less accurate distinguishing high frequencies than low.

- Auditory system filters sounds
  - can attend to sounds over background noise.
  - for example, the cocktail party phenomenon.

3.Touch

- Provides important feedback about environment.
- May be key sense for someone who is visually impaired.
- Stimulus received via receptors in the skin:
  - thermoreceptors – heat and cold
  - nociceptors – pain
  - mechanoreceptors – pressure
    (some instant, some continuous)
- Some areas more sensitive than others e.g. fingers.
- Kinethesis - awareness of body position
  - affects comfort and performance.

4.Movement

- Time taken to respond to stimulus:
  reaction time + movement time
- Movement time dependent on age, fitness etc.
- Reaction time - dependent on stimulus type:
  - visual ~ 200ms
  - auditory ~ 150 ms
  - pain ~ 700ms
- Increasing reaction time decreases accuracy in the unskilled operator but not in the skilled operator.

Movement (cont)

- Fitts' Law describes the time taken to hit a screen target:

$$Mt = a + b \log_2(D/S + 1)$$

where: a and b are empirically determined constants

Mt is movement time

D is Distance

S is Size of target

⇒ targets as large as possible
distances as small as possible

⇒ **In design Focus:** Most frequently used option can be placed closest to the user's start point in the menu list

HUMAN MEMORY

Sensory memory

Short term memory

Long term memory

What is memory?

- It is the ability to store and retrieve the information
- Much of our everyday activity relies on memory
- We need to understand some of the capabilities and limitations of human memory to answer these
  - How does memory works?
  - How do we remember some arbitrary list ?
  - Why do some people remember more easily than others?

Memory

There are three types of memory function:

Sensory memories

Attention

Short-term memory or working memory

Rehearsal

Long-term memory

Selection of stimuli governed by level of arousal.

Sensory memory

- It act as "buffers" for stimuli received through senses
- A sensory memory exists for each sensory channel
  - iconic memory: visual stimulus
  - echoic memory: acoustic stimulus
  - haptic memory: touch stimulus
- Continuously overwritten by new information coming in on these channels

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- Continuously overwritten by new information coming in on these channels
- Examples
  - Iconic memory: At firework displays where moving "sparklers" leave a persistence image, 0.5 sec
  - Echoic memory: brief "play back" of information

Sensory memory

- Information is passed from sensory memory into short-term memory by attention
- Attention is the concentration of the mind on one out of a number of competing stimuli or thoughts
- It is clear we are able to focus our attention to one thing at a time
  - This is due to the limited capacity of our sensory memory
  - Otherwise overloaded

Sensory memory

- We can choose which stimuli to attend to, and this choice is our level of interest or need
- This explains the "cocktail party phenomenon"
  - "We can attend to one conversation over the background noise, but we may choose to switch our attention to a conversation across the room if we here our name mentioned"
- Information received by sensory memories is quickly passed into more permanent or overwritten and lost

Short-term memory (STM)

- It act as a "Scratch-pad" for temporary recall of information
- Examples:
  - Calculate the multiplication 35x6 in your head
  - Comprehensive test
- rapid access ~ 70ms
- rapid decay ~ 200ms

Short-term memory (STM)

- STM also has limited capacity- 7 ± 2 chunks
- There are two methods for measuring memory capacity
  - Recall the sequence in order
  - Recall the sequence in any order

Try this!

Try this!

212348278493202

Try this!

212348278493202

01 21 414 2626

Short-term memory (STM)

- The successful formation of a chunk is known as "closure"
- In Design Focus: Cashing in
  - ATM machine

Short Term Memory (STM):

- In design Focus: 7± 2 revisited
  - List, menu and groups of items should be no more than 7 items long
  - Command line interfaces

Short-term memory (STM)

- Patterns can be useful as aids to memory

Try this!

- Patterns can be useful as aids to memory
  - HEC ATR ANU PTH ETR EET

Try this!

- Patterns can be useful as aids to memory
  - HEC ATR ANU PTH ETR EET

- The sequence is easy to recall (the cat ran up the tree)

Short-term memory (STM):

The "recency effect"

- Recall of the last words presented is better than recall of those in the middle.
- This is known as the "recency effect"

Long-term memory (LTM)

- Repository for all our knowledge
  - slow access ~ 1/10 second
  - slow decay, if any
  - huge or unlimited capacity

- Long-term memory is intended for the long-term storage of information

Long-term memory (LTM)

- There are three main activities in LTM
  - Storage or remembering of information
  - Forgetting
  - Information retrieval

LTM - Storage

- Information is placed in LTM from STM (working memory) though "rehearsal"
- This process can be optimized in a number of ways

LTM - Storage

- Ebbinghaus performed numerous experiments on memory, using himself as a subject.
- He discovered that "amount learned was directly proportional to the amount of time spent in learning"
  - this is known as total time hypothesis
    - "amount retained proportional to rehearsal time"

LTM - Storage

- Baddeley and others suggest that learning time is most effective if it is distributed over time.
  - This is known as distribution of practice effect
    - optimized by spreading learning over time
- However, rehearsal is not enough to learn information well.
- If the information is not meaningful it is more difficult to remember

Try this!

- List A:
  - Faith, Logic, Value,, Large, Age, Path, Magic, Success, Huge, Precision

Try this!

- List B:
  - Boat, Tree, Toy, Pen, Plate, Lamp, Wood, Cycle, Table, Clock

Tips-for Learning

- Structure, meaning and familiarity
  - information easier to remember if it follows the above

Structure- alignment - names

- Usually scanning for surnames make it easy! ⇒

Umarani Jayaraman
Masilamani V
Noor Mahammad
Sadagopan N
Sivaselvan B

Jayaraman , Umarani
V, Masilamani
Mahammad, Noor
N, Sadagopan
B, Sivaselvan

Umarani	Jayaraman
Masilamani	V
Noor	Mahammad
Sadagopan	N
Sivaselvan	B

LTM - Forgetting

- There are two main theories of forgetting
  - **Decay and interference**
- Decay
  - Information held in LTM may eventually be forgotten
  - Information is lost gradually but very slowly
  - **Jost's law** which follows from this, states that "if two memory traces are equally strong at a given time the older one will be more durable"

LTM - Forgetting

- Interference
  - New information replaces old: **retroactive interference**
  - Old may interfere with new: **proactive inhibition**

- Forgetting is also affected by **emotional factors**

LTM - Information retrieval

- Recall
  - Information reproduced from memory
  - It can be assisted by cues, e.g. categories, imagery

- Recognition
  - The presentation of information provides the knowledge that the information has been seen before
  - less complex than recall - information is cue

Try this!

Try this!

READ
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Try this! (Categories are easier to remember)

Tips to Improve your memory

- 1. bun
- 2. shoe
- 3. tree
- 4. door
- 5. hive
- 6. sticks
- 7. heaven
- 8. gate
- 9. wine
- 10. hen

Improve your memory (try to visualize it)

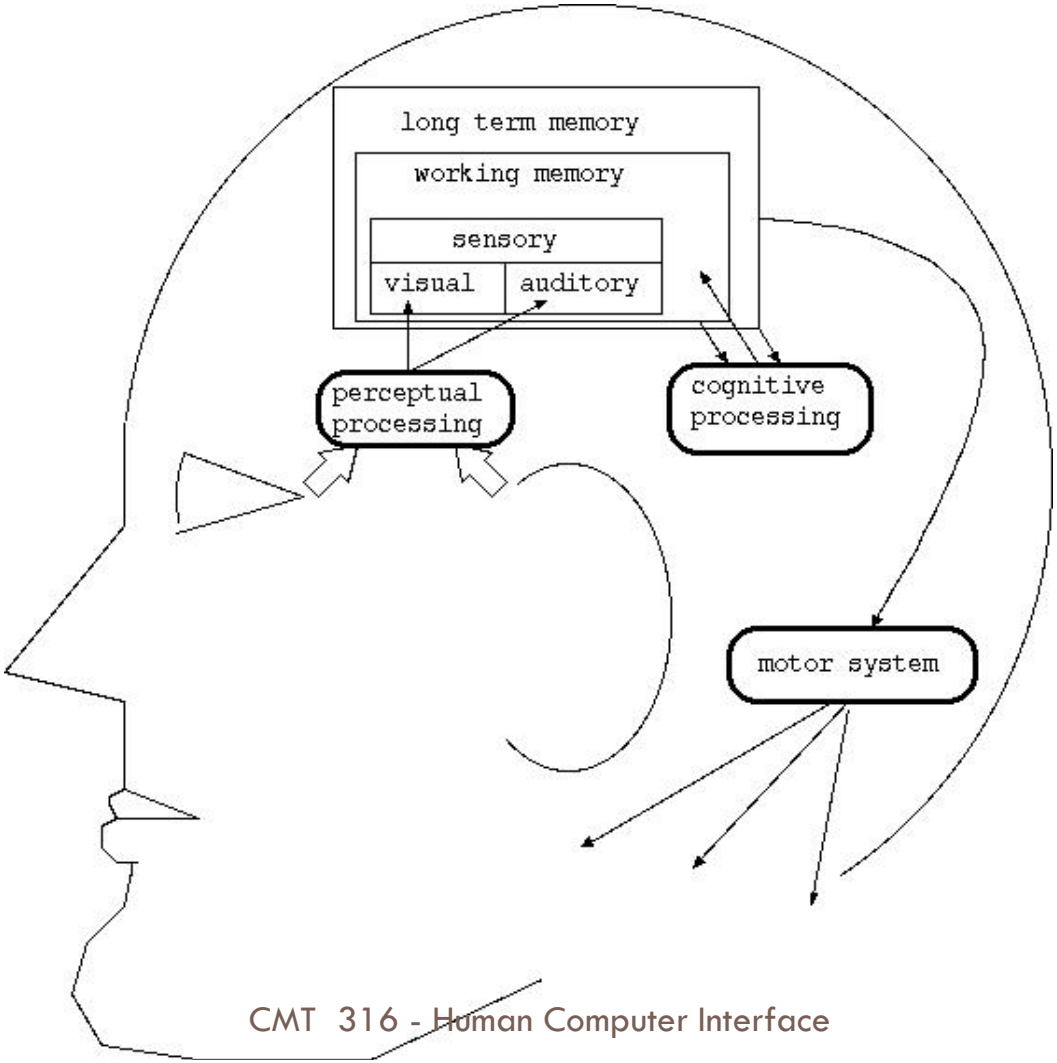
- 1. bun
- 2. shoe
- 3. tree
- 4. door
- 5. hive
- 6. sticks
- 7. heaven
- 8. gate
- 9. wine
- 10. hen

The model human processor (MHP)

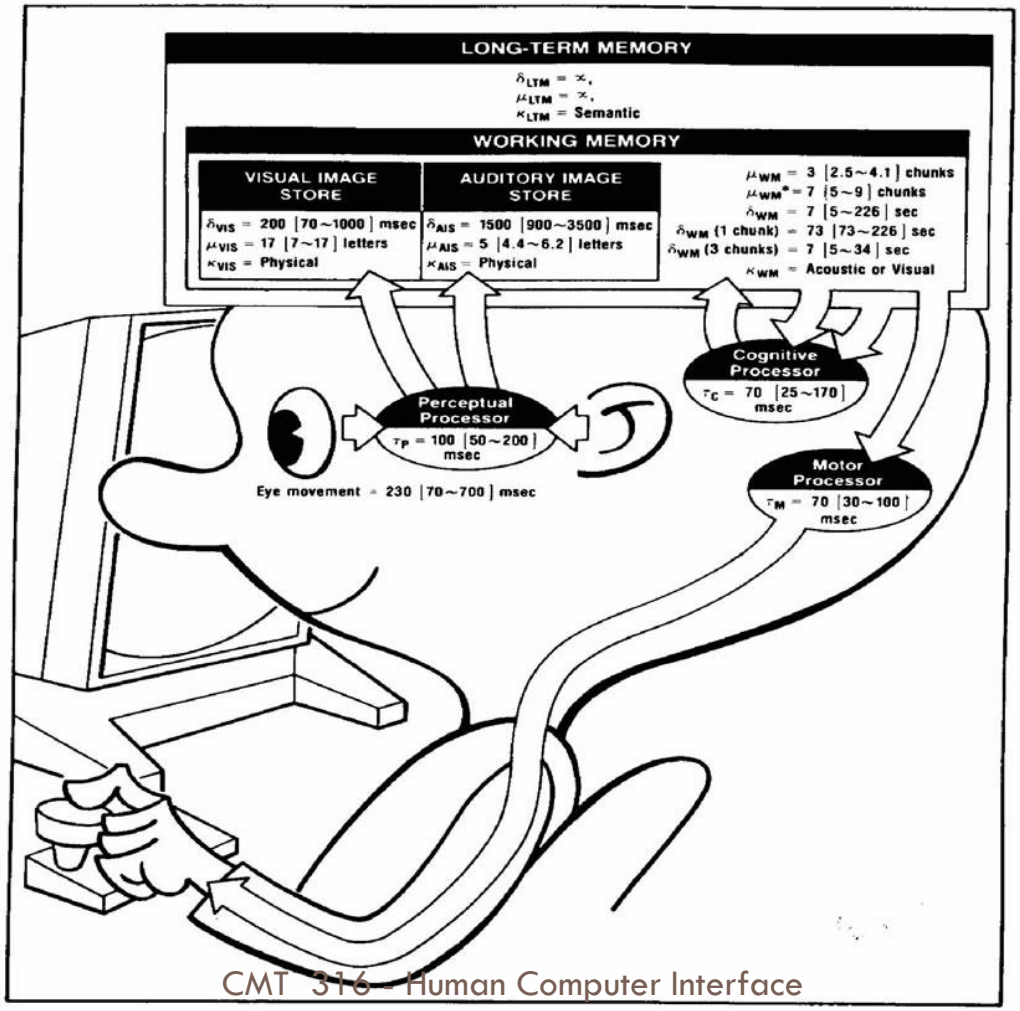
(card, moran and newell, 1983)

- Model human as three interacting subsystems
  - Perceptual system
  - Cognitive system
  - Motor system

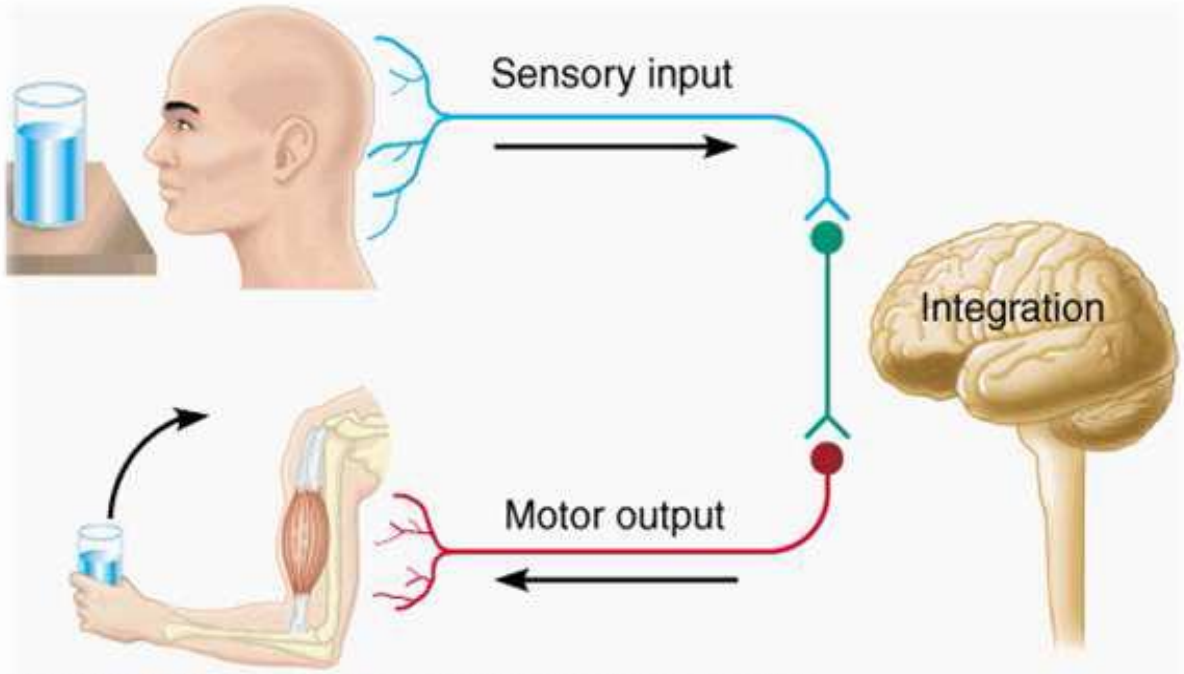
The model human processor



The model human processor



The biology of the model



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